

FRONTEND

PyTorch Model
KAN / MLP

IR Graph

IR GRAPH (6 Operation Types)

MatMul B-spline LUT
Softmax Argmax

Standard Add

BACKEND (Siemens SCL)

S7-1200 Compact DB+FB · 15-pt LUT
S7-1500 Unrolled · 50-pt LUT
DB+FB split for 64 KB block limit

VALIDATION

TIA Portal V21
MCP Openness API
0 errors · 0 warnings
Cross-Validation

OPTIMIZER (6 passes)

Adaptive B-spline LUT
Hoist Binary Search
Dead-Node Elimination

Fuse MatMul Add
LUTize EXP
Strength Reduction

STATIC ANALYZER

Memory Budget (TIA V21 measured)
FLOPs Count · WCET (Z3 SMT)
DA Safety Factor · Per-Category Breakdown

VERIFICATION (3-tier)

Tier 1: Compiler Template Proofs (4/6 op)
Tier 2: Per-Function Z3 Cert. (512/512)
Tier 3: Composition Certificate (~200 lines)